

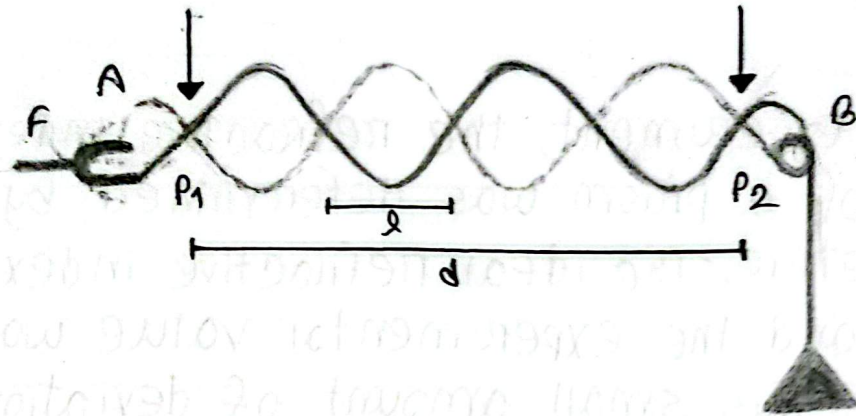


Figure 1: Transverse arrangement

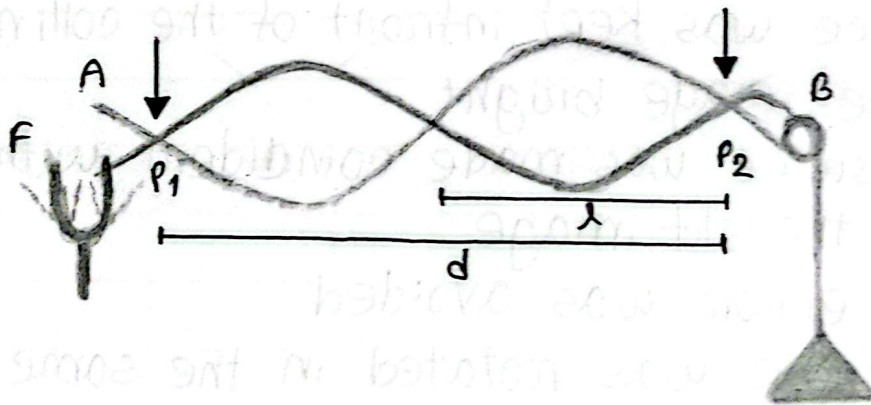


Figure 2: Longitudinal arrangement

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DATE: 04.01.2026

EXPERIMENT OF Determination of the frequency of a tuning fork by Melde's Apparatus

Theory:

In Melde's experiment, one end of a light string is attached to a prong of a tuning fork and the other end passes over a pulley to a scale pan, which provides tension. When the tuning fork is set to vibration, the vibrations may be Transverse or Longitudinal depending on the orientation of the tuning fork.

In the transverse arrangement, the plane of vibration is perpendicular to the string and the frequency of the tuning fork N is equal to frequency of the string n .

On the other hand, in the longitudinal arrangement, the plane of vibration is parallel to the string and the frequency of the tuning fork is twice of the string, $N = 2n$.

Transverse arrangement, $N = n = \frac{1}{\lambda} \sqrt{\frac{T}{m}} \dots (i)$

Longitudinal arrangement, $N = 2n = \frac{2}{\lambda} \sqrt{\frac{T}{m}} \dots (ii)$

Wavelength of the stationary wave, $\lambda = \frac{2l}{p}$

Tension in the string, $T = (w + w_1)g \dots (iii)$



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Required Apparatus:

1. Melde's Apparatus
2. String
3. Weight box
4. Balance

Procedure:

- (i) Melde's Apparatus was set up with a tuning fork, a light string, a pulley and a mass hanger and the string was attached to the prong of the tuning fork while passing over the pulley
- (ii) For the "Transverse arrangement", the tuning fork was clamped so that its prongs vibrated perpendicular to the length of the string, a known mass was added to provide tension and the effecting vibrating length was adjusted until clear stationary waves were formed
- (iii) The number of loops between two fixed points was counted, the distance between these points was measured and the wavelength was calculated
- (iv) For the "Longitudinal arrangement", the tuning fork was rotated so that its prongs vibrated parallel

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to the string and stationary waves were again obtained by adjusting the tension and vibrating length

(v) Observations were taken for different applied masses in both arrangements and the corresponding frequencies of the string were calculated

(vi) The frequency of the tuning fork was determined from the frequency of the string using the appropriate relation for transverse ($N=n$) and longitudinal ($N=2n$) vibrations

Calculations:

For Transverse arrangement,

$$\lambda_1 = \frac{2d_1}{K} = \frac{2}{4} \times 95.1 = 47.55 \text{ cm}$$

$$T_1 = (w + w_1)g = \left(\frac{18.18 + 4}{1000}\right) \times 9.8 \times 10^5$$

$$= 21.73 \times 10^3 \text{ dyne}$$

$$n_1 = \frac{1}{\lambda_1} \sqrt{\frac{T_1}{m}} = \frac{1}{47.55} \times \sqrt{\frac{21.73 \times 10^3}{6.18 \times 10^{-3}}}$$

$$= 39.44 \text{ Hz}$$

$$N_1 = n_1 = 39.44 \text{ Hz}$$

$$\lambda_2 = \frac{2d_2}{K} = \frac{2}{4} \times 97.5 = 48.75 \text{ cm}$$

$$T_2 = (w + w_1)g = \left(\frac{18.18 + 7}{1000}\right) \times 9.8 \times 10^5$$

$$= 24.67 \times 10^3 \text{ dyne}$$

$$n_2 = \frac{1}{\lambda_2} \sqrt{\frac{T_2}{m}} = \frac{1}{48.75} \times \sqrt{\frac{24.67 \times 10^3}{6.18 \times 10^{-3}}}$$

$$= 40.99 \text{ Hz}$$

$$N_2 = n_2 = 40.99 \text{ Hz}$$

$$\lambda_3 = \frac{2d_3}{K} = \frac{2}{4} \times 99.6 = 49.8 \text{ cm}$$

$$T_3 = (w + w_1)g = \left(\frac{18.18 + 10}{1000}\right) \times 9.8 \times 10^5$$

$$= 27.61 \times 10^3 \text{ dyne}$$

$$n_3 = \frac{1}{\lambda_3} \sqrt{\frac{T_3}{m}} = \frac{1}{49.8} \times \sqrt{\frac{27.61 \times 10^3}{6.18 \times 10^{-3}}}$$

$$= 42.44 \text{ Hz}$$

$$N_3 = n_3 = 42.44 \text{ Hz}$$

Now,

$$N = \frac{N_1 + N_2 + N_3}{3} = \frac{39.44 + 40.99 + 42.44}{3}$$

$$= 40.96 \text{ Hz}$$

Miscellaneous data:

1. Mass of scale pan, $w = 18.18 \text{ g}$
2. Length of string, $L = 110.1 \text{ cm}$
3. Mass of string, $M = 0.68 \text{ g}$

For Longitudinal arrangement,

$$\lambda_1 = \frac{2d_1}{K} = \frac{2}{2} \times 92.8 = 92.8 \text{ cm}$$

$$T_1 = (w + w_1)g = \left(\frac{18.18 + 7}{1000}\right) \times 9.8 \times 10^5$$

$$= 24.67 \times 10^3 \text{ dyne}$$

$$n_1 = \frac{1}{\lambda_1} \sqrt{\frac{T_1}{m}} = \frac{1}{92.8} \times \sqrt{\frac{24.67 \times 10^3}{6.18 \times 10^{-3}}}$$

$$= 21.53 \text{ Hz}$$

$$N_1 = 2n_1 = 2 \times 21.53 = 43.06 \text{ Hz}$$

$$\lambda_2 = \frac{2d_2}{K} = \frac{2}{2} \times 86.5 = 86.5 \text{ cm}$$

$$T_2 = (w + w_1)g = \left(\frac{18.18 + 5}{1000}\right) \times 9.8 \times 10^5$$

$$= 22.72 \times 10^3 \text{ dyne}$$

$$n_2 = \frac{1}{\lambda_2} \sqrt{\frac{T_2}{m}} = \frac{1}{86.5} \times \sqrt{\frac{22.72 \times 10^3}{6.18 \times 10^{-3}}}$$

$$= 22.17 \text{ Hz}$$

$$N_2 = 2n_2 = 2 \times 22.17 = 44.34 \text{ Hz}$$

$$\lambda_3 = \frac{2d_3}{K} = \frac{2}{2} \times 95.8 = 95.8 \text{ cm}$$

$$T_3 = (w + w_1)g = \left(\frac{18.18 + 12}{1000}\right) \times 9.8 \times 10^5$$

$$= 29.58 \times 10^3 \text{ dyne}$$

$$n_3 = \frac{1}{\lambda_3} \sqrt{\frac{T_3}{m}} = \frac{1}{95.8} \times \sqrt{\frac{29.58 \times 10^3}{6.18 \times 10^{-3}}}$$

$$= 22.84 \text{ Hz}$$

$$N_3 = 2n_3 = 2 \times 22.84$$

$$= 45.67 \text{ Hz}$$

Now,

$$N = \frac{N_1 + N_2 + N_3}{3} = \frac{43.06 + 44.34 + 45.67}{3}$$

$$= 44.36 \text{ Hz}$$

4. Mass per unit length of the string,
 $m = \frac{M}{L} = 6.18 \times 10^{-3} \text{ gm/cm}$

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Experimental Data:

Table 01 - Data for frequency of the transverse arrangement

S. No.	Mass on the scale pan, w_1 (g)	Tension, $T = (w + w_1)g$ (dynes)	Total no. of loops (K) between the points P_1 and P_2	Distance between the points P_1 and P_2 , d (cm)	Wave length, $\lambda = \frac{2d}{K}$ (cm)	Mass per unit length of the string, m (gm/cm)	Frequency of the string, $n = \frac{1}{\lambda} \sqrt{\frac{T}{m}}$ (Hz)	Frequency of the fork, $N = n$ (Hz)
1	4	21.73×10^3	4	95.1	47.55	6.18×10^{-3}	39.44	39.44
2	7	24.67×10^3	4	97.5	48.75		40.99	40.99
3	10	26.61×10^3	4	99.6	49.8		42.44	42.44

Table 02 - Data for frequency of the longitudinal arrangement

S. No.	Mass on the scale pan, w_1 (g)	Tension, $T = (w + w_1)g$ (dynes)	Total no. of loops (K) between the points P_1 and P_2	Distance between the points P_1 and P_2 , d (cm)	Wave length, $\lambda = \frac{2d}{K}$ (cm)	Mass per unit length of the string, m (gm/cm)	Frequency of the string, $n = \frac{1}{\lambda} \sqrt{\frac{T}{m}}$ (Hz)	Frequency of the fork, $N = 2n$ (Hz)
1	7	24.67×10^3	2	92.8	92.8	6.18×10^{-3}	21.53	43.06
2	5	22.72×10^3	2	86.5	86.5		22.17	44.34
3	12	29.58×10^3	2	95.8	95.8		22.84	45.67

Results:

 $N = 40.96 \text{ Hz}$ (for transverse arrangement) $N = 44.36 \text{ Hz}$ (for longitudinal arrangement)

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Discussion:

In this experiment, stationary waves were produced in both transverse and longitudinal arrangements, and the variations of wave-length with tension was found to follow the relation $\lambda \propto \sqrt{T}$. The frequencies obtained from both arrangements were in close agreement, validating Melde's method, while small errors were due to friction, air damping and measurement inaccuracies.

Precautions:

- Fractional loop was avoided
- The length between the pulley and the scale pan was kept as short as possible to get the correct tension
- The number of loops was increased/decreased by decreasing/increasing the load on the scale pan